Can we rebuild gexlog's gamma walls from ThetaData?

Validation of TD-computed dealer-GEX walls against gexlog's own saved per-strike profile, 77 morning sessions (Apr–Sep 2026). Verdict: the method **tracks gexlog well** — median net R^2 **0.978** — once two input anchors are set to the cash close.

PROFILE FIT (NET GEX)	CALL WALL MATCH	PUT WALL MATCH
0.978 MEDIAN R^2	**88%** EXACT	**89%** EXACT
≥0.90 on 62/74 days · ≥0.80 on 66/74	≤5pt 88% · ≤10pt 88% · median |Δ| 0.0pt	≤5pt 89% · ≤10pt 89% · median |Δ| 0.0pt
	vs published: exact 53% · ≤5pt 57%	vs published: exact 47% · ≤5pt 48%

What this proves

ThetaData **Standard** serves no greeks and no bulk chains, so we compute gamma ourselves (Black–Scholes) from per-strike **open interest** + the option's **EOD price** (to solve implied vol), then form dealer-GEX = Σ OI·γ·100·S^2·0.01. Regressing our per-strike net GEX against gexlog's *own saved profile* tests whether we reproduce their curve — and therefore their walls.

The two fixes that mattered

Input	Was (wrong)	Now (right)	Effect
IV-solve spot	gexlog's premarket **ES** ref (e.g. 7461)	**Cash EOD close** the prices belong to (7408)	fixes call-side IV on gap days
Gamma center	ES ref	**Cash EOD close**	07-24 net R^2 **0.39** → **0.96**

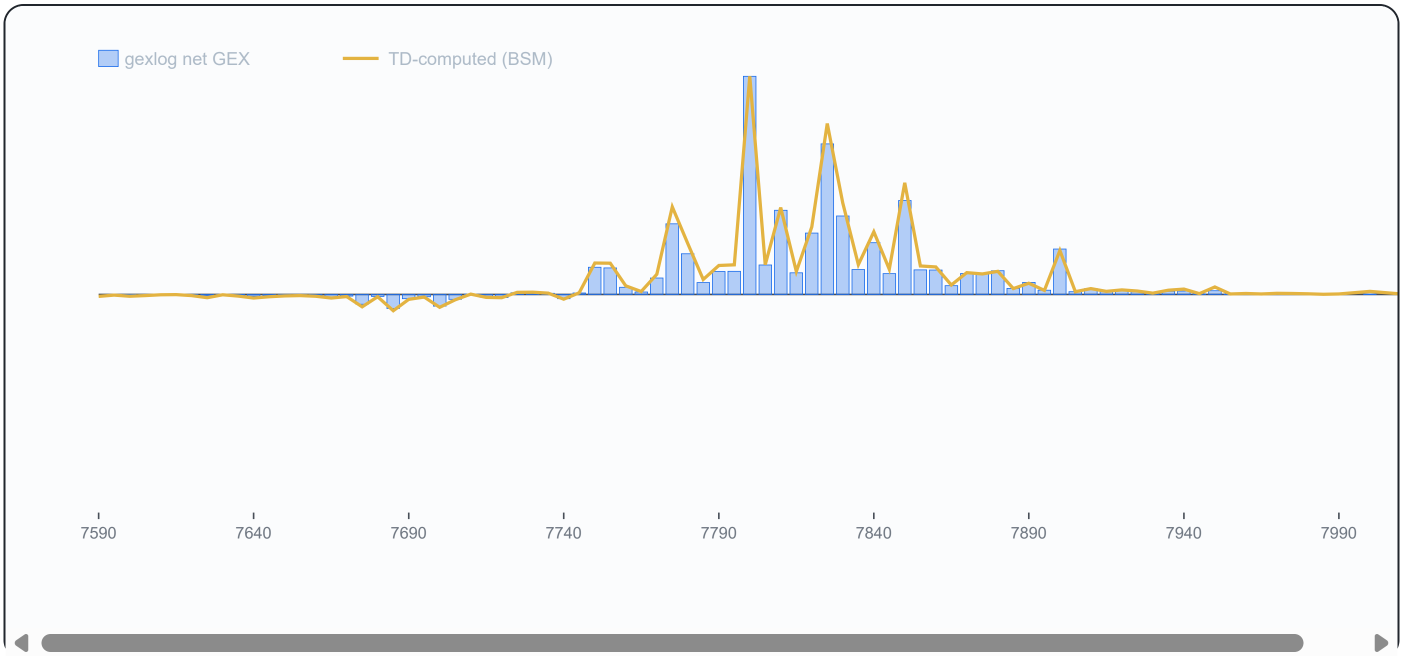
> Both anchor on the **close-to-close cash level** — the same anchor gexlog uses for its Expected Move (verified: EM band centered on the cash close, = cash × VIX%/√252).

Locked method

- **Chain:** 0DTE SPXW (same-day expiry)

- **Open interest:** prior-session close record
- **Implied vol:** per-strike, from the OTM option's prior-close price, solved at the cash close
- **Gamma:** BSM, centered at the cash close
- **GEX:** standard dealer convention (slope ≈ $100 \cdot S^2 \cdot 0.01$, confirmed empirically)
- **Wall:** putWall = max −GEX below spot, callWall = max +GEX above (gexlog's own convention)

Sample overlay — 2026-08-14



Normalised net-GEX by strike: gexlog's saved profile (bars) vs our TD/BSM curve (line).

Per-session results — all 77 days tested

Put wall misses on 8 day(s): median |Δ| 45pt, max 100pt. Call wall misses on 9 day(s): median |Δ| 35pt, max 70pt.

Date	net R^2	TD put/call	gexlog put/call	Δput	Δcall	match
2026-05-13	0.325	7350/7450	7365/7425	-15	+25	
2026-05-14	0.586	7365/7500	7335/7500	+30	0	
2026-05-15	0.702	7375/7560	7375/7560	0	0	✓
2026-05-18	0.191	7300/7450	7350/7475	-50	-25	
2026-05-19	0.981	7360/7500	7360/7500	0	0	✓

Date	net R^2	TD put/call	gexlog put/call	Δput	Δcall	match
2026-05-20	0.994	7340/7450	7340/7450	0	0	✓
2026-05-21	0.948	7375/7435	7375/7475	0	-40	
2026-05-22	0.995	7300/7500	7300/7500	0	0	✓
2026-05-26	—	—	7375/7500	—	—	
2026-05-27	0.989	7450/7530	7450/7530	0	0	✓
2026-05-28	0.968	7475/7530	7475/7530	0	0	✓
2026-05-29	0.949	7485/7600	7485/7600	0	0	✓
2026-06-01	0.985	7475/7600	7475/7600	0	0	✓
2026-06-02	0.961	7550/7615	7550/7615	0	0	✓
2026-06-03	0.989	7555/7630	7555/7630	0	0	✓
2026-06-04	0.866	7500/7625	7540/7625	-40	0	
2026-06-05	0.928	7500/7625	7500/7625	0	0	✓
2026-06-08	0.914	7375/7505	7375/7505	0	0	✓
2026-06-09	0.997	7375/7500	7375/7500	0	0	✓
2026-06-10	0.925	7355/7525	7355/7525	0	0	✓
2026-06-11	0.997	7220/7330	7220/7330	0	0	✓
2026-06-12	0.985	7320/7430	7320/7500	0	-70	
2026-06-15	0.819	—	—	—	0	
2026-06-22	—	—	7500/7575	—	—	
2026-06-23	0.979	7450/7530	7450/7530	0	0	✓
2026-06-24	0.961	7330/7515	7330/7515	0	0	✓
2026-06-25	0.987	7320/7425	7320/7425	0	0	✓
2026-06-26	0.990	7325/7415	7325/7415	0	0	✓
2026-06-29	0.948	7325/7425	7325/7425	0	0	✓
2026-06-30	0.980	7300/7450	7300/7450	0	0	✓
2026-07-01	0.938	7475/7540	7475/7540	0	0	✓
2026-07-02	0.969	7400/7500	7400/7500	0	0	✓

Date	net R^2	TD put/call	gexlog put/call	Δput	Δcall	match
2026-07-06	—	—	7425/7570	—	—	
2026-07-07	0.963	7425/7550	7475/7550	-50	0	
2026-07-08	0.961	7475/7525	7475/7525	0	0	✓
2026-07-09	0.919	7475/7550	7475/7550	0	0	✓
2026-07-10	0.980	7450/7550	7450/7550	0	0	✓
2026-07-13	0.997	7440/7600	7440/7600	0	0	✓
2026-07-14	0.993	7475/7550	7475/7550	0	0	✓
2026-07-15	0.994	7535/7600	7535/7600	0	0	✓
2026-07-16	0.887	7555/7600	7555/7600	0	0	✓
2026-07-17	0.692	7520/7580	7520/7550	0	+30	
2026-07-20	0.997	7450/7565	7450/7565	0	0	✓
2026-07-21	0.981	7410/7540	7410/7540	0	0	✓
2026-07-22	0.998	7450/7550	7450/7550	0	0	✓
2026-07-23	0.991	7485/7545	7485/7545	0	0	✓
2026-07-24	0.960	7400/7470	7300/7470	+100	0	
2026-07-27	0.975	7400/7500	7400/7500	0	0	✓
2026-07-28	0.986	7400/7525	7400/7525	0	0	✓
2026-07-29	0.986	7300/7490	7300/7490	0	0	✓
2026-07-30	0.980	7300/7450	7300/7450	0	0	✓
2026-07-31	0.970	7400/7500	7400/7500	0	0	✓
2026-08-03	0.958	7350/7530	7350/7530	0	0	✓
2026-08-04	0.419	7475/7630	7525/7630	-50	0	
2026-08-05	0.959	7565/7750	7565/7750	0	0	✓
2026-08-06	0.092	7675/7750	7700/7770	-25	-20	
2026-08-07	0.995	7650/7750	7650/7750	0	0	✓
2026-08-10	0.980	7600/7775	7600/7775	0	0	✓
2026-08-11	0.985	7730/7775	7730/7775	0	0	✓

Date	net R^2	TD put/call	gexlog put/call	Δput	Δcall	match
2026-08-12	**0.917**	7670/7800	7670/7800	0	0	✓
2026-08-13	**0.978**	7660/7760	7660/7760	0	0	✓
2026-08-14	**0.992**	7685/7800	7685/7800	0	0	✓
2026-08-17	**0.974**	7775/7850	7775/7850	0	0	✓
2026-08-18	**0.982**	7725/7790	7725/7790	0	0	✓
2026-08-19	**0.982**	7630/7725	7630/7725	0	0	✓
2026-08-20	**0.985**	7650/7755	7650/7755	0	0	✓
2026-08-21	**0.679**	7640/7750	7640/7700	0	**+50**	
2026-08-24	**0.997**	7630/7725	7630/7725	0	0	✓
2026-08-25	**0.991**	7600/7700	7600/7665	0	**+35**	
2026-08-26	**0.977**	7600/7700	7600/7700	0	0	✓
2026-08-27	**0.862**	7675/7725	7675/7725	0	0	✓
2026-08-28	**0.977**	7650/7800	7650/7800	0	0	✓
2026-08-31	**0.981**	7650/7750	7650/7750	0	0	✓
2026-09-01	**0.984**	7675/7710	7675/7710	0	0	✓
2026-09-02	**0.978**	7550/7680	7550/7680	0	0	✓
2026-09-03	**0.973**	7550/7680	7550/7680	0	0	✓
2026-09-04	**0.971**	7650/7750	7650/7800	0	**-50**	

Δ = TD wall − gexlog wall, in SPX points (0 = exact strike match; blank match = both walls exact).

Caveat — the residual is vendor OI

After the two fixes, the remaining disagreement is the **irreducible difference between ThetaData's open interest and gexlog's (tradier) feed** — not a modeling error. On most days the OI distributions agree (R^2>0.9); on a few (e.g. 2026-05-13) they diverge enough to move a wall by a strike or two. This is the expected limit of reproducing one vendor's walls from another's data, and it is why we validate rather than assume.